INCH-POUND

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MILITARY SPECIFICATION

LATHE, MULTIPLE PURPOSE, CHUCKING, HORIZONTAL SPINDLE, COMPUTER NUMERICALLY CONTROLLED (CNC)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers numerically controlled lathes of a horizontal spindle, chucking type, capable of performing multiple machining operations on the face and periphery of a workpiece in one chucking. The various operations performed include contour turning and boring, off-center and peripheral hole drilling and milling operations using linear and rotary axes of motion with accessories and equipment specified herein.

1.2 Classification. The lathe shall be of the following sizes. The sizes to be furnished shall be as specified (see 3.5 and 6.2).

Size	Chuck diameter in inches
0	8
1	10
2	11
3	12

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Industrial Plant Equipment Center, ATTN: DIPEC-SSM, Memphis, Tennessee 38114-5051, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 3416

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specification, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATION

MILITARY

MIL-M-18058 - Machinery, Metal and Woodworking, Packaging of.

STANDARDS

FEDERAL

FED-STD-H28 - Screw Thread Standards for Federal Services.

FED-STD-376 - Preferred Metric Units for General Use by the Federal Government.

MILITARY

DoD-STD-100C- Engineering Drawing Practices.

MIL-STD-461 - Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interferences.

MIL-STD-462 - Electromagnetic Interference Characteristics, Measurement of.

(Unless otherwise specified, copies of Federal and military specifications, standards, and handbooks are available from Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

2.1.2 Other Government documents and publications. The following other Government document forms a part of this specification to the extent specified herein. The issues shall be those in effect on the date of the solicitation and may include title changes.

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CODE OF FEDERAL REGULATIONS (CFR)

U. S. DEPARTMENT OF LABOR, OCCUPATIONAL SAFETY AND HEALTH ADMINISTRATION (OSHA)

29 CFR 1910 - Occupational Safety and Health Standards.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402-0001.)

(Copies of specifications, standards, and publications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Non-Government publications. The following documents form a part of this specification to the extent specified herein. The issue of these documents shall be the latest revision of the document on the date of the solicitation and may include title changes.

AMERICAN GEAR MANUFACTURERS' ASSOCIATION (AGMA)

AGMA 360.02 - Manual for Machine Tool Gearing.

AGMA 390.03 - Gear Classification, Materials and Measuring Methods for Unassembled Gears.

(Application for copies should be addressed to the American Gear Manufacturers' Association, Standards Department, 1500 King Street, Suite 201, Alexandria, VA 22314-2717.)

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI B5.9 - Spindle Nose For Toolroom Lathes, Engine Lathes, Turret Lathes and Automatic Lathes (ISO 702).

ANSI B11.6 - Safety Requirements for the Construction, Care, and Use of Lathes.

ANSI B93.114M - Pneumatic Fluid Power, Systems Standard for Industrial Machinery.

ANSI/ISO 54 - Cylindrical Gears For General Engineering and For Heavy Engineering, Modules and Diametral Pitches of.

ANSI/ISO 8860/1 - Dimensional, Physical and Magnetic Characteristics.

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ANSI/ISO 8860/2 - Track Format.

ANSI/ISO 9293 - Volume and File Structure of Flexible Disk Cartridges for Information Interchange.

(Application for copies should be addressed to the American National Standards Institute, ATTN: Sales Dept., 1430 Broadway, New York, NY 10018-3363.)

NATIONAL ELECTRICAL MANUFACTURERS' ASSOCIATION (NEMA)

NEMA MG 1 - Motors and Generators.

NEMA ICS 1 - General Standards for Industrial Controls and Systems.

(Application for copies should be addressed to the National Electrical Manufacturers' Association, 2101 L Street, N.W., Washington, DC 20037-1580.)

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 79 - Electrical Standard for Industrial Machinery.

(Application for copies should be addressed to the National Fire Protection Association, Batterymarch Park, Quincy MA 02269-9101.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D 3951 - Standard Practice for Commercial Packaging.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103-1187.)

ELECTRONIC INDUSTRIES ASSOCIATION (EIA)

RS-227 - One-Inch Perforated Tape.

RS-232 - Interface Between Data Terminal Equipment and Data Communication Equipment Employing Serial Binary Data Interchange.

RS-244 - Character Code for Numerical Machine Control Perforated Tape.

RS-267 - Axis and Motion Nomenclature for Numerically Controlled Machine Tools.

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RS-274 - Interchangeable Variable Block Format for Positioning, Contouring, and Contouring/Positioning Numerically Controlled Machines.

RS-358 - Subset of USA Standard Code for Information Interchange for Numerical Control Perforated Tape.

RS-494 - 32 Bit Binary CL Exchange (BCL) Input Format for Numerically Controlled Machines.

(Applications for copies should be addressed to Electronic Industries Association, Engineering Department, 2001 Eye Street, N.W., Washington, DC 20006-1804.)

(Non-Government standards and other publications are normally available from the organizations which prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets, or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order (see 6.2), a sample shall be subjected to first article inspection (see 4.4 and 6.3).

3.2 Design. The lathe shall be new (not a prototype) and one of the manufacturer's current production models capable of operations in accordance with the requirements herein. The machine shall include all components, parts, and features necessary to meet the performance requirements specified herein. All parts subject to wear, breakage, or distortion shall be accessible for adjustment, replacement, and repair. In addition to performing standard lathe operations, the machine shall perform operations on the face and periphery of the workpiece utilizing the headstock "C" axis and turret mounted rotary tooling. Face and peripheral operations shall include straight and contour milling, reaming, drilling, and tapping.

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3.2.1 Design and construction dimensional systems. Unless otherwise specified (see 6.2), either the U.S. Customary System of Units (US) or the International System of Units (SI) shall be used in the design and construction of the lathe. In this specification, all measurements, dimensions, sizes, and capacities are given in US units. These measurements may be converted to SI units through the use of the conversion factors and methods specified in FED-STD-376.

3.2.2 Measuring and indicating device calibrations. Measuring and indicating devices shall be graduated in US, SI, or dual (US and SI) units as specified by the procuring activity (see 6.2). Regardless of the measurement system used, all measuring and indicating devices on the lathe shall be graduated in the same system.

3.2.3 Energy efficiency. The lathe and its components that directly consume energy in normal operation shall be designed and constructed for energy efficiency as governed by the developments available within the industry.

3.2.4 Controls. All electrical, mechanical, hydraulic, and pneumatic operating controls shall be located convenient to the operator's work station(s).

3.2.5 Safety and health requirements. All machine parts, components, mechanisms, and assemblies furnished on the lathe, whether or not specifically required herein, shall comply with all of the requirements of OSHA 29 CFR 1910 and ANSI B11.6. If a conflict arises between 29 CFR 1910 and ANSI B11.6, 29 CFR 1910 shall apply. Additional safety and health requirements shall be as specified (see 6.2).

3.2.5.1 Safety guarding. Covers, guards, or other safety devices shall be provided for the point-of-operation and all other parts of the machine that present safety hazards. The safety devices shall not interfere with the operation of the lathe. The safety devices shall prevent unintentional contact with the guarded part, and shall be removable to facilitate inspection, maintenance, and repair of the parts. The safety devices shall not interfere with the operation of the machine. If specific point-of-operation guarding dependent on workpiece size and configuration is required, the procuring activity shall specify the exact configuration of the guard required (see 6.2).

3.2.6 Mercury restriction. The lathe shall not contain mercury or mercury compounds nor be exposed to free mercury during manufacture.

3.2.7 Asbestos restriction. Asbestos and materials containing asbestos shall not be used on or in the lathe.

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3.2.8 Environmental protection. Under the operating, service, transportation, and storage conditions described herein, the lathe shall not emit materials hazardous to the ecological system as prohibited by Federal, state, or local statutes in effect at the point of installation.

3.2.9 Lubrication. Means shall be provided to ensure lubrication for all moving parts. Recirculating lubrication systems shall include a filter which is cleanable or replaceable. Each lubricant reservoir shall have at least a 200 hour capacity and have means for determining fluid level. Means shall be provided to indicate a low-oil condition. Should a low-oil condition, occur the machine shall enter an emergency stop condition upon completion of the current operation. All oil holes, grease fittings, and filler caps shall be accessible.

3.2.10 Interchangeability. All parts subject to replacement shall be manufactured to definite dimensions and tolerances. Replacement parts (except for hand-fitted parts such as ways and boxed bearings) shall be interchangeable without requiring modification. Replacement parts are those which are subject to wear or failure such as gears, bearings, shafts, feed screws, drive belts, pumps, and seals. Electrical replacement parts shall include, but not be limited to, motors, relays, fuses, switches, magnetics, pushbuttons, light bulbs, braking mechanisms, and solid-state devices. Tooling and accessories normally supplied as standard items shall also be considered replacement parts. Foreign-sourced replacement parts shall be interchangeable with domestically supplied components or be readily available from a domestic source.

3.3 Construction. The lathe shall be constructed of parts which are new, without defects, and free of repairs. The structure shall withstand all forces encountered during operation of the lathe to its maximum rating and capacity without distortion.

3.3.1 Castings and forgings. All castings and forgings shall be free of defects, scale, and mismatching. No processes such as welding, peening, plugging, or filling with solder or paste shall be used for reclaiming any defective part.

3.3.2 Fastening devices. All screws, pins, bolts, and other fasteners shall be installed to prevent loosening. Fastening devices subject to removal or adjustment shall not be permanently installed.

3.3.3 Surfaces. All surfaces shall be clean and free of harmful or extraneous materials. All edges shall be either rounded or beveled unless sharpness is required to perform a necessary function. Except as otherwise specified herein, the condition and finish of all surfaces shall be in accordance with the manufacturer's commercial practice.

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3.3.4 Welding, brazing, or soldering. Welding, brazing, or soldering shall be employed only where specified in the original design. None of these operations shall be employed as a repair measure for any defective part.

3.3.5 Painting. Unless otherwise specified (see 6.2), the lathe shall be painted in accordance with the manufacturer's commercial practice.

3.3.6 Threads. All threaded parts, either US or SI, used on the lathe and its related attachments and accessories shall conform to FED-STD-H28 and the applicable "Detailed Standard" section referenced therein.

3.3.7 Gears. All gears, shafts, and pinions of spindle and axis drive trains shall be designed and manufactured of a suitable ferrous material to meet or exceed the requirements of AGMA 360.02 and 390.03 for the US system or ANSI/ISO 54 for the SI system. The gears, shafts, and pinions shall be of proper width and size to transmit full-rated torque and horsepower throughout the speed ranges without failure for the expected service life of the machine. Gears, shafts, and pinions in the drive train shall be hardened and ground steel. Working surface hardness shall be not less than Rockwell C-48.

3.3.8 Electromagnetic interference control. When specified (see 6.2), equipment furnished under this specification shall comply with MIL-STD-461. The equipment and subsystems class and the emission and susceptibility requirements shall be as specified.

3.4 Components. The lathe shall consist of not less than a bed and base, headstock, headstock spindle "C" axis, turret(s), feed mechanism, tailstock, and machine control unit.

3.4.1 Bed and base. The bed shall be of the slanted or vertical construction. The vertical construction shall have longitudinal bedways running horizontally one above the other in a vertical plane. The slanted bed configuration shall have longitudinal bedways running horizontal one above the other at an angle not exceeding 45 degrees from vertical. The bedways shall provide horizontal carriage movement along the longitudinal axis. The surfaces of the bedways that support and guide machine slides shall have a surface hardness of not less than Rockwell C-50 or be of an antifriction type. The base shall have a chip and coolant pan designed for holding chips. Slide way design shall incorporate wipers to keep chips and dirt out of the bearing area. When specified (see 6.2), an automatic chip removal system shall be provided. Means shall be provided for securing and leveling the machine on a foundation.

3.4.2 Headstock. The lathe shall have a headstock to support the spindle and spindle drive mechanism. The headstock and all components located therein shall be capable of utilizing the full machine horsepower without vibration or

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chatter. Unless otherwise specified (see 6.2), the spindle nose shall be of the type specified in table I. The spindle nose shall conform to ANSI B5.9 for the type and size provided. All bearings shall be of a type, size, and precision suitable for the application. Either a splash or pressurized system of lubrication shall be used on rotary components of the spindle drive system.

3.4.3 Spindle "C" axis. The machine shall be provided with a drive system to the spindle independent of the main drive motor. This drive system shall provide "C" axis radial positioning and/or contouring of the spindle in response to programmed commands. A feedback system shall be provided to indicate spindle position in degrees and minutes. Positioning accuracies shall be as specified in table II.

3.4.4 Tool turret. The machine shall be equipped with the turret(s) required to meet the tooling requirements of table I, and the performance requirements of table II for the size machine specified under paragraph 1.2. Regardless of the size machine required, the primary turret shall be capable of securing the quantity of rotary tools specified in table I. When rotary tooling is being used, only the active tool shall be rotating. Unless otherwise specified (see 6.2), speed and power requirements of the rotary tools shall be selectable, and meet the requirements specified in table I. A precision coupling shall be used to index tools into position relative to the workpiece. Indexing shall be accomplished by either programmed commands and manual control.

3.4.5 Feed mechanism. The drive unit and feedback device for each slide unit shall form a close-loop system capable of controlling slide movement in precise increments for accurate feed and positioning. Each carriage and cross-slide shall be driven by an antibacklash, precision, recirculating ball screw and servo motor. Each ball screw shall be covered or shielded to protect the drive units and feedback devices from fluids, chips, and dirt. The response to command signals for modes of departure, feed rates, rapid traverse, and start or stop shall be as required to meet or exceed workpiece accuracies specified herein. The feedback device shall accurately measure each slide movement and send signals to the control unit. The input resolution shall be no greater than 0.0001 inch/0.001mm.

3.4.6 Tailstock. Unless otherwise specified (see 6.2), all machines shall be equipped with a tailstock. The tailstock shall be capable of being positioned and clamped at any point within its range of travel. When specified and fully described by the procuring activity (see 6.2), a programmable tailstock shall be provided. The tailstock shall allow unrestricted use of the full cutting stroke of the turret slide.

3.4.7 Computer numerical control (CNC). The lathe shall have a fully automatic, solid-state, soft-wired, integrated-circuit type, computer numerical control. The machine control unit (MCU) shall provide automatic control of machine functions, operating modes, linear axes movement, spindle

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operation, turret rotation, and other part program directed functions for lathes of the type specified herein. Control axes shall be identified in accordance with EIA Standard RS-267. The control shall have linear and circular interpolation and be capable of simultaneous control of two linear axes. Control features shall include an EIA Standard RS-232 communication port for interfacing the control with peripheral devices, buffer storage, fixed cycles, part program storage, part program and buffer edit capability, and control diagnostics. The MCU shall direct machine functions from part program data stored in memory, manually input or directly from the input media specified herein. The control shall initiate a halt or an error message should a fault condition occur in the control. All necessary executive program routines shall be furnished. The control unit shall conform to NFPA 79. A line voltage of +/- 10 percent from normal shall not adversely affect the numerical control system function. The control shall function in ambient temperatures ranging from 30 to 120 degrees Fahrenheit (F) and humidity ranging from 5 to 95 percent noncondensing. The control shall automatically shut down when internal operating temperature exceeds safe operating temperature. The control shall be inch/metric switchable with control resolution of 0.0001 inch/0.001mm for each axis.

3.4.8 Data input.

3.4.8.1 Punched tape data input. When specified (see 6.2), the input media for the control shall be 1-inch, 8-channel punched tape. The data input format and codes shall comply with EIA Standards RS-227, RS-274, and RS-358. When specified (see 6.2), EIA Standard RS-244 shall be used in lieu of EIA Standard RS-358. The control shall be equipped with a photoelectric tape reader capable of holding, feeding, and reading tape prepared in the format specified at a rate not less than 300 characters per second. Unless otherwise specified (see 6.2), the tape reader shall be supplied with standard feed and take-up spools 5-1/4 inches in diameter. The tape reader shall be housed in a dust-free area of the numerical control cabinet and shall be easily accessible for cleaning and tape handling. The control shall have a tape error detection capability to continuously check for the following errors:

a. Odd parity per EIA Standard RS-244.
b. Even parity per EIA Standard RS-358.
c. Misaligned sprocket holes.
d. No sprocket holes.

Any of the above errors shall halt machine functions and illuminate an error light or a message on the data display screen to indicate a fault condition.

3.4.8.2 Binary cutter location. When 32 Bit Binary Cutter Location (BCL) Exchange Input for Numerically Controlled Machines is specified (see 6.2), the input media shall be a 3-1/2-inch hardshell diskette structured in accordance with ANSI/ISO 8860/1, ANSI/ISO 8860/2, and ANSI/ISO 9293. The input data format shall comply with EIA Standard RS-494. Manufacturers providing functions or features that cannot be addressed by command codes currently covered by EIA Standard RS-494 shall apply to the Electronic Industries Association, IE-31 Committee on Numerical Control Systems and Equipment for an extension of the standard codes to cover the new features or functions. The control shall be equipped with a 3-1/2-inch hardshell diskette reader. The control shall have

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the capability to perform a cyclic redundancy check of the input data and be capable of reading not less than 15,000 characters per second.

3.4.8.3 Diskette input. When specified (see 6.2), a portable diskette reader/writer shall be provided with all the hardware and software necessary to interface between the control and reader/writer. The reader/writer shall receive and store part program data from the control unit and be capable of loading stored programs to the memory of the control unit. Part program storage shall be on a 3-1/2-inch hardshell diskette. The diskette shall meet the requirements of ANSI/ISO 8860/1, ANSI/ISO 8860/2, and ANSI/ISO 9293.

3.4.8.4 Manual data input. A means for manual data input (MDI) shall be furnished to permit manual entry of alphabetical, numerical, and special characters necessary for control of all programmable machine functions. The manually input data shall be immediately displayed on the data display to permit data verification. The MDI shall be capable of entering a block of data into memory upon command.

3.4.8.5 Postprocessor. When specified (see 6.2), the system shall be furnished with a postprocessor program(s) to be utilized in off-line computer assisted preparation of part programs. The postprocessor shall be provided for converting specified programming language (see 6.2) to the form and format required to operate the machine tool and control. The postprocessor shall be written for use on the type computer specified (see 6.2). Documentation sufficient to implement and utilize the postprocessor shall be provided to the using activity for computer compatibility verification and validation at least 90 days prior to delivery of the lathe.

3.4.9 Standard control features.

3.4.9.1 Block delete. The block delete feature shall permit bypassing any blocks of information identified by a block delete code for use when a portion of a program is to be omitted.

3.4.9.2 Buffer storage. The MCU shall have buffer storage capable of storing one or more blocks of program data, read either from tape, diskette, or memory. As blocks are transferred to active command, new blocks shall be read into the buffer.

3.4.9.3 Display. Unless otherwise specified (see 6.2), the display shall be a cathode ray tube (CRT). The CRT shall be capable of displaying machine system parameters and program codes from memory, buffer storage, and active data. The CRT shall be capable of displaying 680 characters on a 9-inch CRT or when specified (see 6.2), 1998 characters on a 14-inch diagonal CRT.

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3.4.9.4 Tool offsets. Control shall have not less than 32 pairs of program selectable tool offsets. The range of the offsets shall be not less than +20.0000 inches.

3.4.9.5 Program data edit. Part program edit capability shall be provided as means for editing tape and MDI input to optimize part program data. It shall be possible to modify part program data by adding, deleting, correcting, or completely changing stored data and operating with the edited program.

3.4.9.6 Sequence number search. The operator shall be able to search through the part program data for a designated block number by initiating the search through the use of the MDI keyboard. The search may be conducted through the tape data or stored part program data, depending on which is active.

3.4.9.7 Jog. The jog control shall permit movement of the slide in the plus or minus direction of travel at preset jog rates such as high, low, or selectable rates of jog.

3.4.9.8 Diagnostic capability. The control shall include a diagnostic system for monitoring the control during operation and for locating malfunctions when troubleshooting. It shall bring the system to a controlled stop if a malfunction occurs during operation. The system shall give fault data in understandable code or message form on the display unit which will identify a problem at board level in the control and identify program errors. When specified (see 6.2), the control shall include an artificial intelligence feature in order to provide interactive diagnostics and the capability of making routine operational decisions at the machine tool.

3.4.9.9 Feedrate override. Means shall be provided to override the programmed feedrate from 0 to not less than 120 percent.

3.4.9.10 Spindle speed override. A spindle speed override shall be provided to allow manual adjustment of spindle speed in either continuous adjustment of in steps of not greater than 10 percent of programmed speed. The spindle speed override function shall control not less than the range of override between 60 and 100 percent of programmed speed. The spindle speed override control shall have a position that allows return of speed to that which it was programmed.

3.4.9.11 Programmed and optional stops. The control shall recognize programmed stopping points in the machining cycle. All controlled motions shall be stopped when the code for a programmed stop is read. The optional stop code shall be ignored unless the control is set to recognize the command.

3.4.9.12 Thread cutting. Single point thread cutting shall be provided for producing all standard straight and taper threads, multiple lead threads, and variable lead threads. Both internal and external threading shall be possible. Unless otherwise specified (see 6.2), the machine shall be capable of providing, by means of parts program, any thread pitch of 0.9999 inch or less, and any thread length of the capacity of the machine and less.

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3.4.9.13 Feed hold. The feed hold function shall permit the operator to interrupt feed rate and stop all motion. While in the feed hold condition, the machine may be operated manually. When normal operation is resumed, motions will resume at their previous feed rate.

3.4.9.14 Zero reference. Means shall be provided to establish a zero point at any position throughout the range of departure of each controlled axis. An operator control shall be provided to return the machine slides to the initial zero position for synchronizing the machine with the control.

3.4.9.15 Incremental feed. Incremental feed circuitry shall provide means for manually adjusting each slide position. Movements in the plus and minus directions shall be selectable in increments of 0.0001, 0.001, 0.01, 0.1 and 1.0 inch on each axis. The incremental feed shall function in each mode of operation except the automatic mode.

3.4.9.16 Constant surface speed. The constant surface cutting speed control shall permit maintaining a constant cutting speed, within the limits of the machine, regardless of part diameter. Spindle speed control shall be based on the part diameter and the programmed surface speed to maintain a constant cutting rate.

3.4.9.17 Feedrate programming. Feedrate shall be programmed in either inches per minutes (IPM) or inches per revolution (IPR). In the feed per revolution mode all axial feed movement shall be a function of the spindle rotation. Feed per minute shall be programmed directly in inches per minute.

3.4.9.18 Part program storage. Part program storage shall be in either solid-state, hard-disk, or bubble-type memory. The control unit shall be capable of storing in memory the content of not less than 1000 feet of part program tape.

3.4.10 Optional control unit features. The control unit shall be equipped with the optional features specified in 3.4.10.1 through 3.4.10.8. (See 3.9 for optional lathe attachments.)

3.4.10.1 Diameter programming. When specified (see 6.2), the lathe shall have diameter programming capable of accepting input information dimensioned in diameter rather than radii.

3.4.10.2 Decimal point programming. When specified (see 6.2), the lathe shall accept decimal point programming and shall recognize, and interpret, numbers in a part program that contains a decimal point.

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3.4.10.3 Dwell. If the control is required to provide a dwell period (see 6.2), the desired dwell time shall be programmed using a preparatory code. The programmable dwell time period shall be 1.0 second or less to 99.999 seconds or more.

3.4.10.4 Independent feed back. If independent feedback measuring system is required to be furnished (see 6.2), the feedback measuring system shall function as an independent unit so that accuracy is not affected by other components. It shall be of a type in which there is no physical contact between stationary and moving elements of the measuring device.

3.4.10.5 Computer graphics. When specified (see 6.2), computer graphic capabilities shall be provided as required, and fully described by the procuring activity.

3.4.10.6 Conversational programming. When specified (see 6.2), conversational programming shall be provided as required, and fully described by the procuring activity.

3.4.10.7 Probing capabilities. When specified (see 6.2), the lathe shall have probing capabilities as required, and fully described by the procuring activity.

3.4.11 Coolant system. The coolant system shall include a sump or reservoir, a power driven pump, and all necessary piping. The sump or reservoir shall have sufficient capacity to permit full flow of coolant. The system shall have means for draining and cleaning and shall include a baffle or a strainer which is easily removable for cleaning. Means shall be provided to permit the operator to direct and control the amount of coolant to the cutting tool over the entire work area.

3.4.12 Pneumatic system. When specified (see 6.2), a pneumatic system shall be provided. The pneumatic system shall meet the requirements of ANSI B93.114M. The system shall have an air line filter, water trap, and oil mist lubricator, as necessary. The system shall include all devices such as pressure regulators, pressure reducers, solenoid valves, and check valves that are necessary to meet the requirements specified herein. Air pressure gauges shall be provided to indicate input air pressure in all lines that are downstream from a pressure control device.

3.4.13 Hydraulic system. The hydraulic system shall be complete, including all pumps, valves, piping, cylinders, and pressure controls. Overpressure protection shall be provided in the high pressure line to prevent damage to components. A filter system shall be provided to ensure delivery of clean fluid. The hydraulic reservoir shall have means for determining fluid level, a provision for draining, and a cleanout access plate if the reservoir is not

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removable. The filtering elements shall be easily replaceable or easily cleaned. The reservoir shall have an easily readable positive fluid level gauge. A low lube level will automatically inhibit axes movement until the reservoir is replenished.

3.4.14 Electrical system. The electrical system shall conform to ANSI/NFPA 79. Unless otherwise specified (see 6.2), each machine shall draw all of its electrical power from a single 230/460-volt, 3-phase, 60-Hertz (Hz) circuit having a fused safety disconnect switch or circuit breaker. The lathe shall be initially wired for operation on 230/460 volts.

3.4.14.1 Motors. Motors shall be rated for 1 hour duty and shall have ball or roller bearings of the sealed and permanently lubricated type. Unless otherwise specified (see 6.2), alternating-current (AC) motors shall be designed to operate on 60-Hz and each motor enclosure shall meet the requirements for a drip-proof enclosure. All motors shall meet the requirements of NEMA MG-1. Each motor shall have an identification plate to identify manufacturer, model identification, serial number, voltage, amperage, horsepower, phase, and frequency. Motor horsepower shall be the manufacturer's standard or standard optional horsepower as indicated on the manufacturer's machine design and construction drawing(s).

3.4.14.2 Worklight. Unless otherwise specified (see 6.2), a worklight shall be provided. The light shall have a protective shield, an on-off switch, and shall be adjustable to reflect maximum light on the work area. If a built-in lighting system is provided, the adjustable light described above shall not be required.

3.4.14.3 Hour meter. When specified (see 6.2), the lathe shall be equipped with an hour meter. The hour meter shall be installed to display accumulated operating time of the main drive motor. The meter shall be of the nonresetting type and shall have a range of 0 to 9,999 hours in increments of not greater than 1 hour. The meter shall be sealed to prevent the entrance of dust and moisture and shall be mounted to withstand shock and vibration generated by the lathe.

3.5 Size and capacity. Unless otherwise specified (see 6.2), the size and capacity of the lathe shall be not less than specified in table I for the size specified.

3.6 Alignment tolerances. Each lathe shall meet the alignment tolerances specified in table II when tested using the procedures given in 4.8.2.

3.7 Performance. The lathe shall be capable of meeting the performance requirements of table II, using test procedures given in paragraph 4.8.3.

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3.7.1 Additional performance requirements. Performance features required in addition to, or in lieu of, the requirements herein shall be as specified and fully described (see 6.2).

3.8 Standard equipment. All standard equipment normally required to properly operate the lathe (such as special tools) shall be furnished.

3.9 Optional equipment. Optional equipment shall be furnished as specified and fully described (see 6.2).

3.10 Repair parts. Required repair parts shall be furnished as specified (see 6.2). All repair parts shall be assigned a part number identified in a repair parts manual.

3.11 Marking on instruments, control panels, and plates. All words on plates shall be in the English language. Characters shall be engraved, etched, embossed, or stamped in boldface on a contrasting background. All plates shall be corrosion resistant.

3.11.1 Lubrication plate. A lubrication plate shall be permanently and securely attached to each machine. Unless otherwise specified (see 6.2), the plate shall contain the following information:

Points of lubricant application
Servicing interval
Type of lubricant
Viscosity
Military or Federal specification number for each lubricant
Size and type of each lubricant filter

3.11.2 Nameplate. A nameplate shall be securely attached to each machine. Unless otherwise specified (see 6.2), the nameplate shall contain the information listed below. If the machine is a special model, the model designation shall include the model number of the basic standard machine and a suffix identified in the manufacturer's permanent records.

Nomenclature
Manufacturer's name
Manufacturer's model designation
Manufacturer's serial number
Power input (volts, total amps, phase, frequency)
Contract Number or Order Number
National Stock Number or Plant Equipment Code
Date of manufacture

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3.12 Technical data. When technical data is required, it shall be furnished in accordance with the requirements of DD Form 1423. All technical data furnished shall be written in the English language. All drawings shall be in accordance with DoD-STD-100C and NEMA ICS 1.

3.13 Workmanship. Workmanship of the lathe shall meet all requirements specified herein and shall be of a quality equal to that prevailing among manufacturers producing equipment of the type covered by this specification.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own continental United States facility or any other continental United States facility suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements (see 6.5).

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspections set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).

b. Quality conformance inspection (see 4.5).

c. Acceptance test (see 4.6).

4.3 Inspection conditions. Unless otherwise specified (see 6.2), all inspections, tests, and examinations shall be performed in an indoor facility with ambient conditions of 55 to 100°F and 10 to 95 percent relative humidity.

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4.4 First article inspection. When a first article inspection is required, it shall be applied to the first article submitted in accordance with 3.1. Unless otherwise specified (see 6.2), first article inspection shall consist of the examination in 4.7 and all tests in 4.8. The machine shall pass the first article examination and all tests to be accepted.

4.5 Quality conformance inspection. Quality conformance inspection shall be applied to each item prior to being offered for acceptance under the contract. Unless otherwise specified (see 6.2), quality conformance inspection shall consist of the examination in 4.7, the tests in 4.8, and the inspection in 4.9. The machine shall pass the examination, all tests, and the inspection to be accepted.

4.6 Acceptance test. Acceptance test shall be performed on the lathe to ensure conformance with this specification. Unless otherwise specified (see 6.2), the acceptance test shall be performed only after the lathe is installed at its final location. The acceptance test shall consist of the examination in 4.7 and all tests in 4.8. The machine shall pass the examination and all tests to be accepted.

4.7 Examination. The lathe shall be examined to determine compliance with all requirements of this specification.

4.8 Tests.

4.8.1 Operational test. The lathe shall be operated at no load for not less than four hours. Proper operation of all controls, motors, adjusting mechanisms, and accessories shall be verified during the trial period.

4.8.2 Alignment tolerance test. Each machine shall be tested to determine compliance with the alignment tolerance requirements of 3.6. The alignment tolerance test shall be performed prior to and subsequent to the performance test(s) given in 4.8.3 through 4.8.5. Any machine adjustments required to comply with 3.6 during or after the performance test shall be cause for rejection.

4.8.3 Performance accuracy tests. The machine shall be operated under CNC in each mode of operation to test the machine's response to command data. The test shall be conducted in accordance with the supplier's standard test procedure to check control functions and machine response to commands for slide motions, turret indexing, spindle speed changes, feed rate changes and auxiliary functions. The safety limit stops shall be checked by attempting to exceed the limits of travel of the carriage and cross slide. All disks, instruments, materials, and tools required to perform and evaluate the tests shall be furnished by the supplier.

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4.8.4 Performance test. Two identical parts shall be machined to test accuracies and surface finish capabilities. The material shall be SAE 1045 or equivalent. Tooling design and geometry shall be of the manufacturer's choice providing carbide inserts and negative rake tool holders are used. Spindle speeds, depth-of-cut, and feed rates shall be appropriate to achieve the cubic stock removal rate specified in table II for the size machine required. The test shall include a variety of cuts to include straight turning, taper turning, shoulder and face cuts, arcs, milling and drilling and single point threading. Surface finish on all cuts shall be not greater than 60 microinches. The test pieces shall meet the accuracies of table II. At the option of the procuring agency, each machine shall be subjected to tests in accordance with specific requirements designated by the activity in lieu of or in addition to the stated test (see 6.2).

4.8.5 Electromagnetic interference control tests. Equipment requiring electromagnetic interference control shall be tested for compliance with 3.3.8 using the procedures given in MIL-STD-462.

4.9 Packaging inspection. Packaging of each item shall be inspected to determine compliance with the requirements of section 5.

5. PACKAGING

5.1 Preservation, packing, and marking. Unless otherwise specified, preservation, packing, and marking shall be in accordance with ASTM D 3951. When specified (see 6.2), level A preservation, level A or level B packing, and marking shall be accomplished in accordance with MIL-M-18058.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. This document is intended to be used in the procurement of multiple purpose machines capable of operations not common to lathes such as milling and drilling.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Size required (see 1.2).

c. First article, if required (see 3.1).

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d. If the lathe is required to be configured in a specific measurement system (US or SI), state required system (see 3.2.1).

e. System of units for measuring and indicating devices (see 3.2.2).

f. Additional safety and health requirements, if required (see 3.2.5).

g. Specific safety guarding, if required (see 3.2.5.1).

h. Painting, if different (see 3.3.5).

i. Electromagnetic interference control, if required (see 3.3.8).

j. Chip removal system, if required (see 3.4.1).

k. Spindle nose, if different (see 3.4.2).

l. Rotary tool spindle speeds,if different (see 3.4.4).

m. Specify if tailstock is not required (see 3.4.6).

n. Programmable tailstock, if required (see 3.4.6).

o. Specify punched tape input, if required (see 3.4.8.1).

p. Specify EIA Standard RS-244 in lieu of EIA Standard RS-358, if required (see 3.4.8.1).

q. Specify take-up reels, if different (see 3.4.8.1).

r. Specify binary cutter location, if required (see 3.4.8.2).

s. Specify diskette input, if required (see 3.4.8.3).

t. Postprocessor, if required (see 3.4.8.5).

u. Specify program language, if postprocessor is required (see 3.4.8.5).

v. Specify computer type, if postprocessor is required (see 3.4.8.5).

w. Specify display, if different (see 3.4.9.3).

x. Fourteen-inch diagonal CRT, if required (see 3.4.9.3).

y. Artificial intelligence, if required (see 3.4.9.8).

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z. Thread cutting, if different (see 3.4.9.12).

aa. Diameter programming, if required (see 3.4.10.1).

bb. Decimal point programming, if required (see 3.4.10.2).

cc. Specify dwell period, if different (see 3.4.10.3).

dd. Independent feedback device, if required (see 3.4.10.4).

ee. Computer graphics, if required (see 3.4.10.5).

ff. Conversational programming, if required (see 3.4.10.6).

gg. Probing capabilities, if required (see 3.4.10.7).

hh. Pneumatic system, if required (see 3.4.12).

ii. Electrical system, if different (see 3.4.14).

jj. AC motors, if different (see 3.4.14.1).

kk. Worklight, if different (see 3.4.14.2).

ll. Hour meter, if different (see 3.4.14.3).

mm. Size and capacity, if different (see 3.5).

nn. Additional performance features, if required (see 3.7.1).

oo. Optional equipment, if required (see 3.9).

pp. Repair parts, if required (see 3.10).

qq. Lubrication plate, if different (see 3.11.1).

rr. Nameplate, if different (see 3.11.2).

ss. Inspection conditions, if different (see 4.3).

tt. First article, if different (see 4.4).

uu. Quality conformance, if different (see 4.5).

vv. Acceptance test, if different (see 4.6).

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ww. Performance test, if different (see 4.8.4).

xx. If preservation, packing, and marking in accordance with MIL-M-18058 is required, specify level required and specification number (see 5.1).

6.2.1 Consideration of data requirements. The following data should be considered when this specification is applied on a contract. The applicable Data Item Descriptions (DIDs) should be reviewed in conjunction with the specific acquisition to ensure that only essential data are requested/provided and that the DIDs are tailored to reflect the requirements of the specific acquisition. To ensure correct contractual application of the data requirements, a Contract Data Requirements List (DD Form 1423) must be prepared to obtain the data, except where DoD FAR Supplement 27.475-1 exempts the requirement for a DD Form 1423.

Reference Paragraph	DID Number	DID Title	Suggested Tailoring
.	.	.	.
.	.	.	.

The above DIDs were those cleared as of the date of this specification. The current issue of DOD 5010.12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL), must be researched to ensure that only current, cleared DIDs are cited on the DD Form 1423.

6.2.1.1 Technical data. All technical data ordered using a DD Form 1423 should be written in the English language.

6.3 First article. When first article inspection is required, the contracting officer should provide specific guidance to offerers whether the item(s) should be a preproduction sample, a first article sample, a first production item, a sample selected from the first ___ production items, a production item from the contractor's current inventory (see 3.__), and the number of items to be tested as specified in 4.4. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.4 Changes from previous issues. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

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6.5 Inspection facilities. It is recommended the contract requires all inspections be performed within the limits of the continental United States. The contracting officer shall request each response to a solicitation includes the location where inspection and testing are to be performed.

6.6 Training. Training required by the procuring activity shall be provided as specified in the contract.

6.7 Warranty. Warranty requirements shall be as specified by the procuring activity in the contract.

6.8 Subject term (keyword) listing.

C-axis
CNC
Lathe
Rotary tooling
Slant
Vertical

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TABLE I - Sizes and capacities.

Characteristics	Size 0	Size 1	Size 2	Size 3
Chuck size:	8	10"	12"	15"
Optional sizes	--	11	--	18", 21"
Swing over cross-slide/bedway	14/21	10/20	21	18/29
Headstock spindle speed range RPM	50-5000	32-3500	20-2500	23-3000
Slide travel: Z-axis	23"	24"	42"	44"
X-axis	5.5	12"	14"	12"
Feed rates IPM: Z-axis	0-350	.1-200	0-120	.1-200
X-axis	0-350	.1-200	0-120	.1-200
"C" axis positioning, degrees	360	360	360	360
Turret live tooling positions	5	12	6	12
Total tool positions	10	12	12	12
Rotary tool spindle speed RPM	0-6000	40-4000	10-1500	40-4000
Motor horsepower rotary tools	6	5	4	5
Headstock motor horsepower	15	20	30	30
Optional motor horsepower		10, 15, 30	15, 20	
Tailstock quill diameter	4	3	4	4
Spindle nose size and type	A2-6"	A2-6"	A2-6"	A2-8"
Optional spindle nose			A2-8"	A2-11"

NOTE: The travel distances and speed ranges shall be not less than stated in table I for the size machine ordered, nor greater than the next larger machine presented in table I. The acceptability of offers exceeding the size requirement for the size machine specified shall be at the discretion of the procuring activity.

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TABLE II - Performance accuracy.

Test to be performed	Requirement
Positioning:	
Slide positioning accuracy:	
Z-axis	0.00061
X-axis	0.00032
Slide positioning repeatability:	
Z-axis	0.0006
X-axis	0.00015
Headstock spindle positioning accuracy:	
accuracy in 360 degrees "C" axis	± 15 arc seconds
repeatability in 360 degrees "C" axis	± 10 arc seconds
Tool turret indexing accuracy:	± 5 arc seconds
Alignment:	
Slide alignment Z and X axis parallel and perpendicular to headstock spindle axis	0.0005/ft.
Headstock spindle axis and turret face or tool spindle axis alignment	0.0005/ft.
Headstock spindle runout on periphery	0.0005 TIR
Rotary tooling spindle runout	0.0005 TIR
Turning accuracy:	
Turned diameters cylindrical	0.0008/ft.
Turned diameters round	0.0003 TIR
Facing cut flatness in 10"	0.0005
Maximum dimensional deviation of two parts	0.001
Surface finish not rougher than 60 microinches	
Cubic inch stock removal rate:	
Size 1	20 in^3/min.
Size 2	25 in^3/min.
Size 3	35 in^3/min.

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Custodians:
Army - AL
Navy -SH
Air Force -99

Review activities:
Air Force -84
DLA-GS

User activities:
Army -AR
Navy - YD,MC

Preparing activity:
DLA - IP

Project (3416-0230)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-L-80245A	Lathe, Multiple Purpose, Chucking, Horizontal Spindle, Computer Numcerically Controlled (CNC)

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: –

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

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